

OpenStax Calculus Volume 3

Section 2.3 Exercises: The Dot Product

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For the following exercises, the vectors $\mathbf{u}$ and $\mathbf{v}$ are given. Calculate the dot product $\mathbf{u} \cdot \mathbf{v}$.

1. $\mathbf{u} = \langle 3, 0 \rangle$, $\mathbf{v} = \langle 2, 2 \rangle$
2. $\mathbf{u} = \langle 3, -4 \rangle$, $\mathbf{v} = \langle 4, 3 \rangle$
3. $\mathbf{u} = \langle 2, 2, -1 \rangle$, $\mathbf{v} = \langle -1, 2, 2 \rangle$
4. $\mathbf{u} = \langle 4, 5, -6 \rangle$, $\mathbf{v} = \langle 0, -2, -3 \rangle$

For the following exercises, the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are given. Determine the vectors $(\mathbf{a} \cdot \mathbf{b})\mathbf{c}$ and $(\mathbf{a} \cdot \mathbf{c})\mathbf{b}$. Express the vectors in component form.

5. $\mathbf{a} = \langle 2, 0, -3 \rangle$, $\mathbf{b} = \langle -4, -7, 1 \rangle$, $\mathbf{c} = \langle 1, 1, -1 \rangle$
6. $\mathbf{a} = \langle 0, 1, 2 \rangle$, $\mathbf{b} = \langle -1, 0, 1 \rangle$, $\mathbf{c} = \langle 1, 0, -1 \rangle$
7. $\mathbf{a} = \mathbf{i} + \mathbf{j}$, $\mathbf{b} = \mathbf{i} - \mathbf{k}$, $\mathbf{c} = \mathbf{i} - 2\mathbf{k}$
8. $\mathbf{a} = \mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{b} = \mathbf{j} + 3\mathbf{k}$, $\mathbf{c} = -\mathbf{i} + 2\mathbf{j} - 4\mathbf{k}$

For the following exercises, the two-dimensional vectors $\mathbf{a}$ and $\mathbf{b}$ are given.

(a) Find the measure of the angle θ between $\mathbf{a}$ and $\mathbf{b}$. Express the answer in radians rounded to two decimal places, if it is not possible to express it exactly.

(b) Is θ an acute angle?

9. [T] $\mathbf{a} = \langle 3, -1 \rangle$, $\mathbf{b} = \langle -4, 0 \rangle$
10. [T] $\mathbf{a} = \langle 2, 1 \rangle$, $\mathbf{b} = \langle -1, 3 \rangle$
11. $\mathbf{u} = 3\mathbf{i}$, $\mathbf{v} = 4\mathbf{i} + 4\mathbf{j}$
12. $\mathbf{u} = 5\mathbf{i}$, $\mathbf{v} = -6\mathbf{i} + 6\mathbf{j}$

For the following exercises, find the measure of the angle between the three-dimensional vectors $\mathbf{a}$ and $\mathbf{b}$. Express the answer in radians rounded to two decimal places, if it is not possible to express it exactly.

13. $\mathbf{a} = \langle 3, -1, 2 \rangle$, $\mathbf{b} = \langle 1, -1, -2 \rangle$

14. $\mathbf{a} = \langle 0, -1, -3 \rangle$, $\mathbf{b} = \langle 2, 3, -1 \rangle$
15. $\mathbf{a} = \mathbf{i} + \mathbf{j}$, $\mathbf{b} = \mathbf{j} - \mathbf{k}$
16. $\mathbf{a} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\mathbf{b} = \mathbf{i} + \mathbf{j} - 2\mathbf{k}$
17. [T] $\mathbf{a} = 3\mathbf{i} - \mathbf{j} - 2\mathbf{k}$, $\mathbf{b} = \mathbf{v} + \mathbf{w}$, where $\mathbf{v} = -2\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$ and $\mathbf{w} = \mathbf{i} + 2\mathbf{k}$
18. [T] $\mathbf{a} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, $\mathbf{b} = \mathbf{v} - \mathbf{w}$, where $\mathbf{v} = 2\mathbf{i} + \mathbf{j} + 4\mathbf{k}$ and $\mathbf{w} = 6\mathbf{i} + \mathbf{j} + 2\mathbf{k}$

For the following exercises determine whether the given vectors are orthogonal.

19. $\mathbf{a} = \langle x, y \rangle$, $\mathbf{b} = \langle -y, x \rangle$, where x and y are nonzero real numbers
20. $\mathbf{a} = \langle x, x \rangle$, $\mathbf{b} = \langle -y, y \rangle$, where x and y are nonzero real numbers
21. $\mathbf{a} = 3\mathbf{i} - \mathbf{j} - 2\mathbf{k}$, $\mathbf{b} = -2\mathbf{i} - 3\mathbf{j} + \mathbf{k}$
22. $\mathbf{a} = \mathbf{i} - \mathbf{j}$, $\mathbf{b} = 7\mathbf{i} + 2\mathbf{j} - \mathbf{k}$
23. Find all two-dimensional vectors $\mathbf{a}$ orthogonal to vector $\mathbf{b} = \langle 3, 4 \rangle$. Express the answer in component form.
24. Find all two-dimensional vectors $\mathbf{a}$ orthogonal to vector $\mathbf{b} = \langle 5, -6 \rangle$. Express the answer by using standard unit vectors.
25. Determine all three-dimensional vectors $\mathbf{u}$ orthogonal to vector $\mathbf{v} = \langle 1, 1, 0 \rangle$. Express the answer by using standard unit vectors.
26. Determine all three-dimensional vectors $\mathbf{u}$ orthogonal to vector $\mathbf{v} = \mathbf{i} - \mathbf{j} - \mathbf{k}$. Express the answer in component form.
27. Determine the real number α such that vectors $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j}$ and $\mathbf{b} = 9\mathbf{i} + \alpha\mathbf{j}$ are orthogonal.
28. Determine the real number α such that vectors $\mathbf{a} = -3\mathbf{i} + 2\mathbf{j}$ and $\mathbf{b} = 2\mathbf{i} + \alpha\mathbf{j}$ are orthogonal.
29. [T] Consider the points $P(4, 5)$ and $Q(5, -7)$.
 - (a) Determine vectors $\vec{OP}$ and $\vec{OQ}$. Express the answer by using standard unit vectors.
 - (b) Determine the measure of angle O in triangle OPQ. Express the answer in degrees rounded to two decimal places.
30. [T] Consider points $A(1, 1)$, $B(2, -7)$, and $C(6, 3)$.
 - (a) Determine vectors $\vec{BA}$ and $\vec{BC}$. Express the answer in component form.
 - (b) Determine the measure of angle B in triangle ABC. Express the answer in degrees rounded to two decimal places.
31. Determine the measure of angle A in triangle ABC, where $A(1, 1, 8)$, $B(4, -3, -4)$, and $C(-3, 1, 5)$. Express your answer in degrees rounded to two decimal places.
32. Consider points $P(3, 7, -2)$ and $Q(1, 1, -3)$. Determine the angle between vectors $\vec{OP}$ and $\vec{OQ}$. Express the answer in degrees rounded to two decimal places.

For the following exercises, determine which (if any) pairs of the following vectors are orthogonal.

33. $\mathbf{u} = \langle 3, 7, -2 \rangle$, $\mathbf{v} = \langle 5, -3, -3 \rangle$, $\mathbf{w} = \langle 0, 1, -1 \rangle$
34. $\mathbf{u} = \mathbf{i} - \mathbf{k}$, $\mathbf{v} = 5\mathbf{j} - 5\mathbf{k}$, $\mathbf{w} = 10\mathbf{j}$
35. Use vectors to show that a parallelogram with equal diagonals is a rectangle.
36. Use vectors to show that the diagonals of a rhombus are perpendicular.
37. Show that $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$ is true for any vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$.
38. Verify the identity $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$ for vectors $\mathbf{u} = \langle 1, 0, 4 \rangle$, $\mathbf{v} = \langle -2, 3, 5 \rangle$, and $\mathbf{w} = \langle 4, -2, 6 \rangle$.

For the following problems, the vector $\mathbf{u}$ is given.

- (a) Find the direction cosines for the vector $\mathbf{u}$.
- (b) Find the direction angles for the vector $\mathbf{u}$ expressed in degrees. (Round the answer to the nearest integer.)

39. $\mathbf{u} = \langle 2, 2, 1 \rangle$
40. $\mathbf{u} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$
41. $\mathbf{u} = \langle -1, 5, 2 \rangle$
42. $\mathbf{u} = \langle 2, 3, 4 \rangle$
43. Consider $\mathbf{u} = \langle a, b, c \rangle$ a nonzero three-dimensional vector. Let $\cos \alpha$, $\cos \beta$, and $\cos \gamma$ be the direction cosines of $\mathbf{u}$. Show that $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$.
44. Determine the direction cosines of vector $\mathbf{u} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ and show they satisfy $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$.

For the following exercises, the vectors $\mathbf{u}$ and $\mathbf{v}$ are given.

- (a) Find the vector projection $\mathbf{w} = \text{proj}_{\mathbf{u}} \mathbf{v}$ of vector $\mathbf{v}$ onto vector $\mathbf{u}$. Express your answer in component form.
- (b) Find the scalar projection $\text{comp}_{\mathbf{u}} \mathbf{v}$ of vector $\mathbf{v}$ onto vector $\mathbf{u}$.

45. $\mathbf{u} = 5\mathbf{i} + 2\mathbf{j}$, $\mathbf{v} = 2\mathbf{i} + 3\mathbf{j}$
46. $\mathbf{u} = \langle -4, 7 \rangle$, $\mathbf{v} = \langle 3, 5 \rangle$
47. $\mathbf{u} = 3\mathbf{i} + 2\mathbf{k}$, $\mathbf{v} = 2\mathbf{j} + 4\mathbf{k}$
48. $\mathbf{u} = \langle 4, 4, 0 \rangle$, $\mathbf{v} = \langle 0, 4, 1 \rangle$
49. Consider the vectors $\mathbf{u} = 4\mathbf{i} - 3\mathbf{j}$ and $\mathbf{v} = 3\mathbf{i} + 2\mathbf{j}$.
 - (a) Find the component form of vector $\mathbf{w} = \text{proj}_{\mathbf{u}} \mathbf{v}$ that represents the projection of $\mathbf{v}$ onto $\mathbf{u}$.
 - (b) Write the decomposition $\mathbf{v} = \mathbf{w} + \mathbf{q}$ of vector $\mathbf{v}$ into the orthogonal components $\mathbf{w}$ and $\mathbf{q}$, where $\mathbf{w}$ is the projection of $\mathbf{v}$ onto $\mathbf{u}$ and $\mathbf{q}$ is a vector orthogonal to the direction of $\mathbf{u}$.
50. Consider vectors $\mathbf{u} = 2\mathbf{i} + 4\mathbf{j}$ and $\mathbf{v} = 4\mathbf{j} + 2\mathbf{k}$.
 - (a) Find the component form of vector $\mathbf{w} = \text{proj}_{\mathbf{u}} \mathbf{v}$ that represents the projection of $\mathbf{v}$ onto $\mathbf{u}$.

- (b) Write the decomposition $\mathbf{v} = \mathbf{w} + \mathbf{q}$ of vector $\mathbf{v}$ into the orthogonal components $\mathbf{w}$ and $\mathbf{q}$, where $\mathbf{w}$ is the projection of $\mathbf{v}$ onto $\mathbf{u}$ and $\mathbf{q}$ is a vector orthogonal to the direction of $\mathbf{u}$.
51. A methane molecule has a carbon atom situated at the origin and four hydrogen atoms located at points $P(1, 1, -1)$, $Q(1, -1, 1)$, $R(-1, 1, 1)$, and $S(-1, -1, -1)$ (see figure).
[Figure: This figure is the 3-dimensional coordinate system. There are four points plotted. The first point is labeled "P(1, 1, -1)," the second point is labeled "Q(1, -1, 1)," the third point is labeled "R(-1, 1, 1)," and the fourth point is labeled "S(-1, -1, -1)." There are line segments from Q to P, P to R and R to Q. There are also two vectors in standard position. The first has terminal point of R and the second has terminal point of S. The angle between them is represented with an arc.]
- (a) Find the distance between the hydrogen atoms located at P and R.
- (b) Find the angle between vectors $\vec{OS}$ and $\vec{OR}$ that connect the carbon atom with the hydrogen atoms located at S and R, which is also called the bond angle. Express the answer in degrees rounded to two decimal places.
52. [T] Find the vectors that join the center of a clock to the hours 1:00, 2:00, and 3:00. Assume the clock is circular with a radius of 1 unit.
53. Find the work done by force $\mathbf{F} = \langle 5, 6, -2 \rangle$ (measured in Newtons) that moves a particle from point $P(3, -1, 0)$ to point $Q(2, 3, 1)$ along a straight line (the distance is measured in meters).
54. [T] A sled is pulled by exerting a force of 100 N on a rope that makes an angle of 25° with the horizontal. Find the work done in pulling the sled 40 m. (Round the answer to one decimal place.)
55. [T] A father is pulling his son on a sled at an angle of 20° with the horizontal with a force of 25 lb (see the following image). He pulls the sled in a straight path of 50 ft. How much work was done by the man pulling the sled? (Round the answer to the nearest integer).
[Figure: This figure is an image of a person pulling a child on a sled. The rope for pulling the sled is represented by a vector and labeled "25 lb." There is an angle between the rope vector and the horizontal ground of 20 degrees.]
56. [T] A car is towed using a force of 1600 N. The rope used to pull the car makes an angle of 25° with the horizontal. Find the work done in towing the car 2 km. Express the answer in joules ($1\text{J} = 1\text{N} \cdot \text{m}$) rounded to the nearest integer.
57. [T] A boat sails north aided by a wind blowing in a direction of $\text{N}30^\circ\text{E}$ with a magnitude of 500 lb. How much work is performed by the wind as the boat moves 100 ft? (Round the answer to two decimal places.)
58. Vector $\mathbf{p} = \langle 150, 225, 375 \rangle$ represents the price of certain models of bicycles sold by a bicycle shop. Vector $\mathbf{n} = \langle 10, 7, 9 \rangle$ represents the number of bicycles sold of each model, respectively. Compute the dot product $\mathbf{p} \cdot \mathbf{n}$ and state its meaning.
59. [T] Two forces $\mathbf{F}_1$ and $\mathbf{F}_2$ are represented by vectors with initial points that are at the origin. The first force has a magnitude of 20 lb and the terminal point of the vector is point $P(1, 1, 0)$. The second force has a magnitude of 40 lb and the terminal point of its vector is point $Q(0, 1, 1)$. Let $\mathbf{F}$ be the resultant force of forces $\mathbf{F}_1$ and $\mathbf{F}_2$.
- (a) Find the magnitude of $\mathbf{F}$. (Round the answer to one decimal place.)

- (b) Find the direction angles of $\mathbf{F}$. (Express the answer in degrees rounded to one decimal place.)
60. [T] Consider $\mathbf{r}(t) = \langle \cos t, \sin t, 2t \rangle$ the position vector of a particle at time $t \in [0, 30]$, where the components of $\mathbf{r}$ are expressed in centimeters and time in seconds. Let $\vec{OP}$ be the position vector of the particle after 1 sec.
- (a) Show that all vectors $\vec{PQ}$, where $Q(x, y, z)$ is an arbitrary point, orthogonal to the instantaneous velocity vector $\mathbf{v}(1)$ of the particle after 1 sec, can be expressed as $\vec{PQ} = \langle x - \cos 1, y - \sin 1, z - 2 \rangle$, where $x \sin 1 - y \cos 1 - 2z + 4 = 0$. The set of point Q describes a plane called the normal plane to the path of the particle at point P.
- (b) Use a CAS to visualize the instantaneous velocity vector and the normal plane at point P along with the path of the particle.